perplexity

Principle (Simple Version)

This experiment uses two reactions:

- Sodium carbonate (**Na_2CO_3**) and sodium hydroxide (**NaOH**) are both bases.
- They are each titrated (measured by reaction) with hydrochloric acid (**HCl**), a strong acid.

Na_2CO_3 reacts with **2 HCl** to make salt, water, and carbon dioxide:

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$$

NaOH reacts with HCl to make salt and water:

$$NaOH + HCl \rightarrow NaCl + H_2O$$

By carefully measuring how much HCl it takes to react with known amounts of base, you can figure out the strength (concentration) of each solution.

Procedure (Easy Steps)

1. **Prepare Standard Solution:**
    - Weigh 0.53 g of sodium carbonate (**Na_2CO_3**).
    - Dissolve it in water and make the volume up to 100 ml. Mix well.
2. **Standardize HCl:**
    - Pipette 20 ml of the sodium carbonate solution into a flask.
    - Add a few drops of methyl orange indicator.
    - Titrate with HCl (fill burette with HCl) until the color changes from yellow to orange-red. Record this volume.
    - Repeat until results are close (concordant).
3. **Titrate NaOH:**
    - Pipette 20 ml of the given NaOH solution.
    - Add a few drops of phenolphthalein (solution will be pink).
    - Titrate with the same (now standardized) HCl until the pink color disappears (colorless).
    - Repeat for consistancy.

All steps make sure you can measure accurately how strong the acid and base solutions are, using simple color changes.